

Digital Logic & Design

EE-1005

Sessional-I Exam

Total Time: 1 Hour

Total Marks: 90

Total Questions: 5 Pages: 6

Date: February 26th, 2024

Course Instructor(s)

Dr. Mehwish Hassan, Mr. Muhammad Aamir Gulzar,
Mr. Shams Farooq,

Semester: Spring-2024

Campus: Islamabad

Dept: CS, SE

Student Name

Roll No

Section

Student Signature

Vetted by

Vetter Signature

	Q1	Q2	Q3	Q4	Q5	Total
Marks Obtained						
Total Marks	20	25	10	10	25	90

Question 1 [20 Marks]

- i. $(1032.21)_4 + (23.1301)_4 = (?)_4$
Perform base-4 addition where base 4 has $\{0,1,2,3\}$

$$\begin{array}{r} 1032.2100 \\ + 23.1301 \\ \hline 1123.0001 \end{array}$$

$$\begin{array}{l} 4-4=0 \\ 7-4=2 \\ 6-4=2 \end{array}$$

- ii. $(201)_3 - (1.1210)_3 = (?)_3$
Perform base-3 subtraction where base 3 has $\{0,1,2\}$

$$\begin{array}{r} 201.0000 \\ - 1.1210 \\ \hline 122.0110 \end{array}$$

National University of Computer and Emerging Sciences

FAST School of Computing, Islamabad Campus

iii. $(101.1)_2 * (1.101)_2 = (?)_2 = (?)_8$ Perform base-2 multiplication where base 2 has $\{0,1\}$

$$\begin{array}{r}
 1011 \\
 1101 \\
 \hline
 1011 \\
 0000 \\
 1011 \\
 0111 \\
 \hline
 100011100
 \end{array}$$

$(10.74)_8$

iv. Count in base given below.

Decimal	Base 3	Base 4	Base 5	Decimal	Base 3	Base 4	Base 5
0	P	L	@	11	QQR	NO	% #
1	Q	M	#	12	RRP	OL	% %
2	R	N	%	13	RRQ	OM	% %
3	QP	O	&	14	RRR	ON	% %
4	QQ	ML	*	15	QPPP	OO	% %
5	QR	mm	#	16	QPPQ	MLL	% %
6	RP	mN	#	17	QPPR	MLm	% %
7	RQ	MO	# %	18	QPQP	MLN	% %
8	RR	NL	# %	19	QPQQ	MLO	% %
9	QPP	NM	# %	20	QPQR	mmm	% %
10	QPQ	NN	% %	21	QPRR	mmm	% %

Question 2 [25 Marks]

Convert the following as indicated.

NOTE: All steps are required no marks for direct answer

i. $(10011.111)_2 = (?)_{10} = (?)_{16}$

$$16 \quad 1 \quad 14 \rightarrow E$$

$$13 \quad 0.5$$

$$19 = \frac{1}{2} + 0.25 + 0.125$$

$$= 19.875$$

$$(13.E)_{16}$$

ii. $(DC93)_{16} = (?)_8$

$$\begin{array}{cccc} 1101 & 1100 & 1001 & 0011 \\ \hline 15 & 6 & 2 & 2 & 3 \end{array}$$

iii. $(33.78125)_{10} = (?)_2$

$$= (10001.11001)_2$$

$$.78125 \times 2 = 1.5625$$

$$0.5625 \times 2 = 1.125$$

$$0.125 \times 2 = 0.25$$

$$0.25 \times 2 = 0.5$$

$$0.5 \times 2 = 1$$

iv. $(295)_{BCD} + (86)_{BCD}$

$$\begin{array}{r} 0010 \quad 1001 \quad 0101 \\ 1000 \quad 0110 \\ \hline 0010 \quad 1011 \\ 0110 \quad 0110 \\ \hline 0011 \quad 1000 \quad 0001 \\ 3 \quad 8 \quad 1 \end{array}$$

v. The ten-bit Gray code for $(353)_{10}$ is 0111010001. Explain briefly but precisely why it cannot possibly be true that 0111010000 is the ten-bit Gray code for 354_{10} . Or calculate gray code for $(354)_{10}$

not

$$354 =$$

$$1381 = (1+0+4+10+0+9+10)_2$$

$$354 = (111010011)_2$$

gray

Question 3 [10 Marks]

- i. Perform 2's Complement Subtraction $(-25)_{10} - (245)_{10}$ and verify your answer by converter it back to decimal

①

$$\begin{array}{r}
 111100111 \\
 - 1100001011 \\
 \hline
 10111100101
 \end{array}$$

-270

$0100001110 = -270$

$ \begin{array}{r} 25 \\ - 12 \\ \hline 13 \\ - 6 \\ \hline 7 \\ - 3 \\ \hline 4 \end{array} $	$ \begin{array}{r} 245 \\ - 122 \\ \hline 123 \\ - 61 \\ \hline 62 \\ - 30 \\ \hline 32 \\ - 15 \\ \hline 17 \\ - 7 \\ \hline 10 \\ - 3 \\ \hline 7 \\ - 1 \\ \hline 6 \end{array} $
---	---

$-254 = 1100001011$

- ii. Perform Subtraction $(314)_8 - (402)_8$ using 7's complement method.

$$\begin{array}{r}
 777 \\
 402 \\
 \hline
 375
 \end{array}$$

$375 - (7^3 - 1) \text{ comp of } 402$

$$\begin{array}{r}
 314 \\
 + 376 \\
 \hline
 710
 \end{array}$$

$5 + 4 = 9 - 7 = 2$
 $8 + 7 = 15 - 8 = 7$

No carry means answer is -ve

$$\begin{array}{r}
 777 \\
 710 \\
 \hline
 -067
 \end{array}$$

Question 4 [Marks 10]

i. $[(\overline{AB} + \overline{AC}) + \overline{A} \overline{B} C](\overline{A} + C)$ Simplify using Boolean Algebra

$$= [(\overline{A} + \overline{B}) \cdot (\overline{A} + \overline{C}) + \overline{A} \overline{B} C](\overline{A} + C)$$

$$= [\overline{A} + \overline{B} \overline{C} + \overline{A} \overline{B} C] (\overline{A} + C)$$

$$= [\overline{A} + \overline{B} \overline{C}] (\overline{A} + C)$$

$$= \overline{A} \cdot \overline{A} \overline{C} + \overline{A} \overline{B} \overline{C} \cdot C$$

$$= 0 \cdot \overline{C} + \overline{A} \overline{B} \overline{C}$$

$$= \overline{A} \overline{B} \overline{C}$$

ii. Convert the following expression given below to Minimized SOP expression using Boolean algebra only.

Note: You are supposed to convert expression to Product of Maxterm and extract sum of Minterm

$$(p + q)(p + \overline{q} + z)(\overline{p} + q + \overline{z})$$

$$= (p + q + z) (p + q + \overline{z}) (\overline{p} + q + z) (\overline{p} + q + \overline{z})$$

$$P(A, B, C) (0, 1, 2, 5)$$

$$P(A, B, C) (3, 4, 6, 7)$$

$$\underline{011} + 100 + 110 + 111$$

$$= (\overline{x} y z) + x \overline{y} \overline{z} + x y \overline{z} + x y z$$

$$= \overline{x} y z + x \overline{y} \overline{z} + x y (\overline{z} + z)$$

$$= \overline{x} y z + x \overline{y} \overline{z} + x y$$

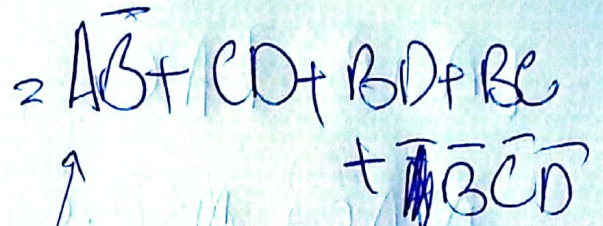
$$= \overline{x} y z + x (\overline{y} \cdot \overline{z} + y)$$

	$\overline{y} \overline{z}$	$\overline{y} z$	$y \overline{z}$	$y z$
$\overline{x}$	0	0	1	0
x	1	1	0	1

$$= \overline{x} y z + x (\overline{y} + \overline{z})$$

$$= \overline{x} y z + x \overline{y} + x \overline{z}$$

1, 1, 6, 0, 0, 1, 6, 0, 0, 0, 0, 1, 0, 0, 0, 0



- $$F(L, M, N, O) = \sum(0, 1, 2, 4, 5, 6, 7, 10, 15)$$



$$= \bar{L}M + \bar{L}\bar{O} + mNO + \bar{m}N\bar{O}$$